# PROJECT STATUS / EXECUTIVE SUMMARY

Objective: create original, device-accurate LineageOS 23.2 (Android 16) bring-up for Redmi Note 17 5G, codename mist, Qualcomm SM4450 / parrot. The sm4450-development organization is only an architectural reference; do not copy another device's partitions, fstab, vendor blobs or boot settings.

## Latest status - important change since prior handover

- LATEST ACTIONS RUN: #43179826520, job #131052829905, stock configuration review, FAILED on 12 Nov 2026 at 08:03 UTC (16:03 Malaysia).
- Exact error: scripts/review_stock_configuration.py line 144 emitted SyntaxError: invalid escape sequence '\'' during python3 -W error::SyntaxWarning -m py_compile. Artifact download succeeded; parse step failed; report publication skipped.
- FIX PUSHED to copilot_project/main: commit 00a7059e29c4f2ee33ed774d27f4481e109809dd, replacing 19 invalid backtick escapes in review script.
- This correction has NOT been retested in a new workflow run. Start the next chat by checking for a newer run; if none exists, dispatch stock-config-review.yml on main.

## Current acceptance state

- Stock ROM acquisition, sparse super unpack and EROFS inventory: verified successful.
- Vendor source archive and seven boot image SHA256 checks: verified successful.
- Detailed VINTF/fstab/init/module report: BLOCKED by the now-fixed, not-yet-rerun Python compile error.
- Production proprietary vendor and kernel-prebuilt trees: still empty. Device tree: six preliminary files, not build-ready.
- No LineageOS 23.2 ROM has been built, booted, or hardware tested. Do not claim any device feature works.

Primary latest run: https://github.com/nobodyAtall-code/copilot_project/actions/runs/43179826520

## Repository map and strict repository rules

Research, staging, workflow tooling and reports:
https://github.com/nobodyAtall-code/copilot_project

Device configuration (six skeleton files):
https://github.com/nobodyAtall-code/android_device_xiaomi_mist

Proprietary blobs and vendor makefiles (empty):
https://github.com/nobodyAtall-code/proprietary_vendor_xiaomi_mist

Stock-compatible kernel boot inputs/prebuilts (empty):
https://github.com/nobodyAtall-code/android_device_xiaomi_mist-kernel

Existing kernel source (Linux 6.6.118):
https://github.com/nobodyAtall-code/android_kernel_xiaomi_mist

## Non-negotiable constraints

- ALL work is committed directly to the existing main branches. No pull requests; no newly created branches.
- Only copilot_project may contain GitHub Actions workflows, extraction, research scripts, temporary archives, experiment logs or publication automation.
- Keep device/vendor/kernel production repositories restricted to required, reviewed build and runtime files. Never bulk copy stock images or HyperOS APKs.
- GH_PAT exists as an Actions secret in copilot_project. Do not reveal or log it. No successful blob-publishing permission test has been shown.
- Do not modify local LineageOS source checkout or remove old sky-related local customizations until inspected and backed up.
- Use stock mist evidence, not unverified sky values from sm4450-development. Check AVB, dynamic partition groups and mount configuration before building/flash.

## Latest repository branch identities

copilot_project main after syntax fix: 00a7059e29c4f2ee33ed774d27f4481e109809dd

android_device_xiaomi_mist main: be0c8b740d8fed1200b4109f13d7c0d208714449

android_kernel_xiaomi_mist main: fe8024a4c5eea6cf1135e615d44e7a3fb15fedac

Vendor and kernel-prebuilt repositories are GitHub-empty and have no main commit yet; contents/branch lookups return 404 for empty repositories.

## Device and source baseline

- Hardware: Redmi Note 17 5G, mist, Qualcomm SM4450 / parrot.
- Target: LineageOS 23.2, Android 16 generation.
- Global HyperOS firmware: OS3.0.304.0.WPUMIXM, Android 16; complete fastboot ROM MD5 c4450c4e168052ca9de1211306d3beff.
- Stock kernel: Linux 6.6.118, mist build config and mist.bzl in existing kernel source.
- Boot header v4; physical boot, init_boot, vendor_boot, dtbo, recovery; EROFS logical partitions; Virtual A/B; F2FS userdata.

## Verified acquisition and archive runs

Run #43172012149: SUCCESS. Downloaded and MD5-verified Global ROM, unpacked eight EROFS logical partitions, indexed 11,313 files and 3,030 shared libraries/symlinks.

Run #43174896778: SUCCESS. Staged research inventories, all shared libraries, stock vendor-source archive and boot-chain images. All four outputs uploaded. Optional raw partition artifact intentionally skipped.

Run #43178657329: SUCCESS. Checked stock vendor archive checksum and hashes for seven boot chain images; inspected 4,373 archive entries; audit artifact contained 360 files. Minor Python backtick warnings in an earlier script fixed later.

## Critical source artifacts

- Run #43174896778 research: artifact 13923423842, 289,196-byte ZIP.
- Run #43174896778 all-so: artifact 13923129711, 443,962,342-byte ZIP.
- Run #43174896778 vendor-source: artifact 13923059799, 466,442,771-byte ZIP.
- Run #43174896778 boot-images: artifact 13924356647, 339,752,592-byte ZIP.
- Run #43178657329 staged-audit report: artifact 13924823795, 278,442-byte ZIP.

All five artifacts were originally configured with 14-day retention and are expected to expire about 26 Nov 2026. Download/promote essential inventories before expiration. Retention is not proof they remain currently available.

## Known stock inventory - what it does and does not prove

- Eight mounted logical images: system_a, system_ext_a, product_a, vendor_a, vendor_dlkm_a, system_dlkm_a, odm_a, mi_ext_a, all EROFS.
- vendor_a indexed 3,129 entries, including HAL libraries, firmware, VINTF fragments, init scripts, and fstab.default/fstab.emmc.
- odm_a indexed 452 entries, including VINTF, init, and permissions files.
- vendor_dlkm_a indexed 278 entries (273 under lib/modules); system_dlkm_a indexed 213 entries (110 under lib/modules).
- Research generated 3,539 vendor/ODM candidate paths and 42 exclusions. Candidates are not an approved LineageOS proprietary-files.txt.
- Extracted logical image file sizes are NOT dynamic-group capacities. Need actual super / lpdump group allocation limits and block-device metadata.

## Recently committed research and workflow history

097928d: stock acquisition and partition extraction audit in research/stock-audit-run-43174896778.md.

230fcfe: archive inventory script scripts/audit_staged_artifacts.py.

7bde4ca and 3df9275: staged-stock-audit.yml and corrected checksum path.

0d73cca: corrected harmless escaped backticks in audit_staged_artifacts.py.

974c1c4: research/staged-archive-audit-run-43178657329.md.

2a48504: scripts/review_stock_configuration.py, structured VINTF/fstab/init/module parser.

6501ac4: stock-config-review.yml, report upload and research-only commit to main.

00a7059: fixed 19 invalid escaped backticks in the review script after run #43179826520 failed.

## Caution about the newly added review workflow

It requests actions:read and contents:write, downloads the successful staged audit artifact, validates input audit.json, compiles and runs the parser, uploads Markdown/JSON reports, then commits research/generated/mist-stock-config.* to copilot_project/main. The last failure occurred BEFORE parser execution; success of the repaired script is still unverified.

## Required execution plan for the next chat

- 01 - FIRST inspect latest copilot_project Actions runs, specifically #43179826520 failure and any runs after the fix 00a7059. Do not assume a fresh run exists.
- 02 - If not already successful, manually dispatch 'Mist stock configuration review' on main: .github/workflows/stock-config-review.yml. Inspect steps, complete logs, artifacts and eventual research/generated Markdown/JSON.
- 03 - If the parser fails again, fix the exact failing source lines in copilot_project/main and rerun. The previous failure was Python warning-as-error caused by invalid '\'' escape.
- 04 - Review generated VINTF HAL entries, actual fstab flags, init services, module load/blocklist/dependencies and generated report for malformed or unsafe data.
- 05 - Build a curated proprietary-files.txt with only essential mist HALs, libraries, firmware and configuration; validate ABI and ELF dependencies. Never blindly copy all candidates or unrelated HyperOS apps.
- 06 - Decode and inspect stock boot/init_boot/vendor_boot/recovery/dtbo; extract kernel release/DTB, ramdisk module order and AVB requirements. Populate only validated kernel-prebuilt inputs.
- 07 - Obtain verified dynamic partition sizes/groups and complete BoardConfig, device.mk, init, fstab, overlays, VINTF, AVB, recovery and SELinux for Android 16.
- 08 - Inventory and back up local LineageOS 23.2 source checkout before selectively removing stale device integration. Then attempt compile, diagnose errors, and only later perform cautious boot tests.

## Potential blockers

- Stock config report not published because latest run failed; fix is pushed but untested.
- No explicit verified production proprietary manifest or extracted full approved vendor set.
- No verified kernel boot packaging, DTB/modules dependency resolution or AVB chain.
- GitHub artifact expiration and size limitations; ensure future binary publisher obeys strict allowlists and only targets main.
- No local source checkout inventory, successful compilation or device test logs.

## Continuation prompt - paste into the next chat

Continue my Redmi Note 17 5G (mist), Qualcomm SM4450/parrot, LineageOS 23.2 Android 16 bring-up using this handover PDF. Use GitHub connector and START by checking copilot_project Actions run 43179826520 and any newer runs. Latest run failed compiling scripts/review_stock_configuration.py due invalid escaped backticks; fix committed to copilot_project/main at 00a7059e29c4f2ee33ed774d27f4481e109809dd. Verify whether stock-config-review.yml succeeded after fix; if not, start the workflow from main, inspect actual errors and fix them. Next inspect research/generated/mist-stock-config.md and .json; use stock data for clean device, vendor and kernel-prebuilt bring-up. Respect main-only/no PR/no new branch; all workflows/research in copilot_project only; no bulk copying HyperOS, sky or private partitions; protect GH_PAT; preserve local Lineage checkout until inventoried/backed up. Do not claim ROM build or boot success without proof.

## Future ’handover’ instruction

Whenever I say ’handover’, produce a fresh, professionally formatted PDF dated for that handover. Check live GitHub main branches, newest Actions runs, failures, artifacts and commits. Include verified work, exact failing errors/fixes, pending items, repository restrictions, risks, links, and a paste-ready next-chat continuation prompt. Provide a downloadable link to the PDF. Do not provide chat-only summary as a substitute.

## Important links

Research hub: https://github.com/nobodyAtall-code/copilot_project

Latest failed run: https://github.com/nobodyAtall-code/copilot_project/actions/runs/43179826520

New workflow:
https://github.com/nobodyAtall-code/copilot_project/actions/workflows/stock-config-review.yml

Successful staged audit:
https://github.com/nobodyAtall-code/copilot_project/actions/runs/43178657329

Successful firmware staging:
https://github.com/nobodyAtall-code/copilot_project/actions/runs/43174896778